

Special Course

Hard Carbon Synthesis & Characterisation for Sodium-Ion Anodes
(BioNa / DTU Energy / DTU Physics)

Project Information

- **Project owner:** BioNa (DTU startup), hosted at DTU Energy
- **Location:** DTU, Kongens Lyngby
- **Academic supervision:** Prof. Johan Hjelm (johh@dtu.dk)
- **Industrial supervision:** Dr. Peter Curran (pcurran@dtu.dk)
- **Day-to-day laboratory supervision:** Dr. Nihat Sahin (negsa@dtu.dk)

Overview

This project presents an opportunity for the student to gain hands-on experience of Hard Carbon production, characterisation, and electrochemical testing. The primary goal is to establish a baseline understanding of the carbonisation process, to provide a foundation for subsequent advanced multi-stage carbonisation protocols. The work supports BioNa's longer-term goal of optimising performance of biowaste derived Hard Carbon anodes in sodium-ion batteries.

Course Description

- Execute pyrolysis experiments at varying furnace temperatures between 600°C and 1300°C.
- Characterize the samples with Yield (mass-out / mass-in), X-Ray Diffraction (extracting d_{002}), Raman spectroscopy (Extracting I_D/I_G), and N_2 - Gas adsorption (BET fitting to extract m^2g^{-1})
- Construct half-cells for selected Hard Carbons in coin-cell format and conduct Galvanostatic Charge/Discharge (GCD) screening to evaluate: Reversible Specific Capacity (mAh/g) & Initial Coulombic Efficiency (%).
- Use the Baseline carbonisation as a reference point to demonstrate a proof-of-principle for advanced "smart temperature profiles" by varying the ramp rate, N_2 flow rate, and introducing dwell points at intermediate temperatures.

Learning Objectives

Students completing the special course will be able to:

1. Safely operate high-temperature furnaces to run controlled pyrolysis processes
2. Perform and interpret characterisation of functional carbons
 - X-Ray Diffraction; Raman Spectroscopy; N_2 - gas adsorption (BET fitting)
3. Receive an introduction to advanced characterisation techniques of Small Angle X-Ray Scattering and He-pycnometry - opportunity to observe collection and interpretation of data
4. Search academic literature for comparison and to give context for results
5. Interpret measurements, storyboard them to form a coherent narrative and ultimately define follow-on campaign(s) of experiments.
6. Under supervision prepare a prototype slurry formulation to coat hard carbons into viable anode sheets.
7. Under supervision construct half-cells in coin format and conduct basic electrochemical testing.
8. Produce an extended written report documenting methods, results, and conclusions

Support and scope

The student undertaking this project will receive in-person support from BioNa employees to increase experimental throughput and remote supervision from senior researchers for analysis

and interpretation. Affords the candidate opportunity to flexibly dictate the direction of project towards their own interests e.g. Cell testing, material synthesis, material characterisation.

Deliverables

- Extended report
- Experiment logbook
- Raw characterisation data (XRD, Raman, BET)
- Physically labelled hard carbon samples handed to BioNa